CIT 140 · Week 7 · Data-Quality Report

Meridian Forge · fictional inspection export v1

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Import status	20 rows loaded

Amber cells: your responses. Other results calculate from Prepared and Audit.

Rule	Check	Rows flagged	Share of all rows	**Affected Row_IDs**
Q01	Machine ID is required	2	10%	R03, R14
Q02	Work order is unique	2	10%	R05, R06
Q03	Status uses the agreed list	2	10%	R08, R16
Q04	Duration is in range	3	15%	R10, R18, R20
Q05	READY identifies an inspector	3	15%	R03, R07,R12
Distinct rows with any issue		11	55%	R03, R05,R06,R07,R08,R10,R12,R14,R16,R18,R20

Scope: all 20 supplied records. Q02 flags every row in a repeated Work_Order_ID group; extra duplicate copies are a different measure. Unresolved fields remain flagged, not guessed.

Source: authored CIT140 case W07-S01 (Raw) and W07-S02 (rules). Report a data-quality finding, not a machine safety clearance. Explain your evidence and next action on page 2.

Week 7 · Explanation and refresh evidence

Finding: Row_ID, Q-number, actual value, source and effect

Row R03 fails Q01 because the Machine_ID is blank. When looking at W07-S02, missing identifiers break tracking, meaning reviewers can't tie this inspection to actual equipment.

Next action: what to verify, who owns it, and how to retest

The Maintenance Records Owner needs to check the paper log for R03's missing ID. Once updated in the raw data, refresh the sheet and re-test Q01.

Limit: why you cannot guess a correction or prove accuracy

Guessing missing values just invents fake data instead of finding the real facts, which means it breaks rule W07-S02. On top of that rules W07-S01 and W07-S03 remind us that passing checks just means the text formatted right and it doesn't actually prove the machine was safe on the floor.

Professional connection: one framework idea and how you used it

I used the Data Quality Framework's "fitness for purpose" concept. Basically, data is only useful if you can trust it to make decisions, so I audited uniqueness and validity before relying on this sheet while following the guide

Refresh test: before, changed value, after, and restored result

Before: Q04 had 3 flagged rows, and R01 had 20 minutes. Changed value: Set R01 to 0 minutes on the Raw sheet. After: Hit Refresh All, and Q04 increased to 4 flagged rows. Restored: Changed R01 back to 20 and refreshed, returning the Q04 count to 3.

Export only Report: two landscape pages, page break before row 26. Review both pages.